Nidhi Ray

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Objective

Computer Engineering student seeking a 2027 internship in hardware, embedded systems, or systems software. VEX V5 team captain, RoboJackets electrical subteam member, and co-author of a published math paper.

Education

Georgia Institute of Technology | Atlanta, GA *August 2026 – Present*
Bachelor of Science in Computer Engineering Expected Graduation: May 2030
Threads: Computing Hardware & Emerging Architectures; Systems & Architecture
John H. Martinson Honors Program (JMHP). Entered with 43 hours of transfer and AP credit; sophomore standing.

SUNY Westchester Community College | Valhalla, NY *June 2024 – August 2026*
Coursework in mathematics, physics, and chemistry, 27 credit hours, GPA 4.00/4.00

Skills

Programming: Python, C++, Java, HTML/CSS, LaTeX
Hardware and Tools: Breadboard prototyping, surface-mount soldering, KiCad, VEX V5 robotics platform, Wolfram Mathematica
Professional Organizations: Society of Women Engineers (SWE), Women in Electrical and Computer Engineering (WECE)
Technical Foundations: Feedback control, discrete mathematics, stochastic processes
Languages: English (fluent), Bengali (native), Spanish (intermediate)

Projects

RoboJackets, RoboCup Team | Electrical Subteam | Georgia Tech *September 2026 – Present*
- Designing a printed circuit board around the Raspberry Pi RP2040 microcontroller in KiCad, working in a small group from schematic capture through board layout.
- Coordinating pinout and connector interfaces so all groups' boards daisy-chain into one system.

DC-DC Converter Workshop | HIVE Makerspace, Georgia Tech *September 2026*
- Built a linear DC-DC converter on a breadboard and measured its power conversion efficiency with a digital storage oscilloscope, DC power supply, and digital multimeter.

VEX V5 Competitive Robotics | Sacred Heart Greenwich *September 2023 – June 2026*
Team Captain
- Programmed autonomous and driver-control routines in C++ on the VEX V5 Brain using motor encoder and sensor feedback.

Published Mathematics Research | Remote *August 2020 – August 2023*
- Co-authored *"Distance-dependent chase-escape on trees,"* in the peer-reviewed New Zealand Journal of Mathematics (2024) with Professor Matthew Junge of Baruch College, deriving necessary and sufficient conditions for species coexistence in a parasite-host growth process. doi.org/10.53733/342

Relevant Coursework

In Progress, Fall 2026: Introduction to Computing (Python); Differential Equations; ECE Discovery Practicum
Planned, Spring 2027: Digital System Design; Introduction to Object-Oriented Programming
Completed: Linear Algebra; Multivariable Calculus; Calculus I and II; Inorganic Chemistry I with lab; Engineering Physics I and II

Activities

Canada/USA Mathcamp | Selected Participant *July 2024 and July 2025*
- Selected for a competitive residential mathematics program: graph theory, group theory, network algorithms.

Manhattan School of Music Precollege and Georgia Tech | Drummer *September 2021 – Present*
- Completed the four-year Precollege program (2021–2025): audition-based, full-day Saturday conservatory study.
- Earned the program certificate, Dean's List, and a New York State School Music Association All-State score of 99.
- Sole drummer for the Georgia Tech student rock band Soon, performing at public campus events since 2026.

Honors and Awards

- AIME qualifier 2020, 2021, and 2022 (top 5 percent of AMC 12, top 2.5 percent of AMC 10 nationally).
- Math Prize for Girls at MIT, top 300 nationally (2020, 2022); 8th of 120, Girls in Math at Yale (2023).